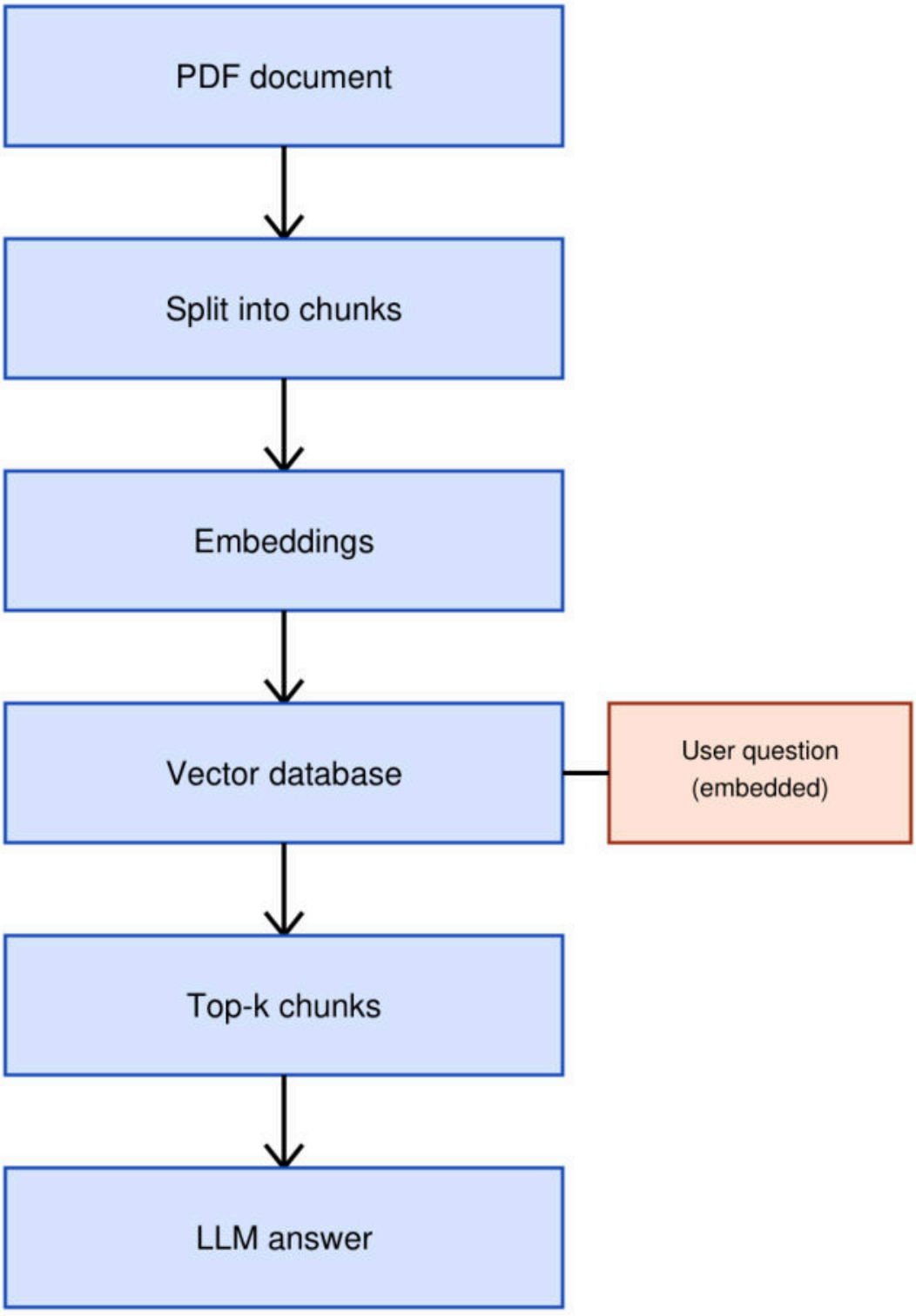
Lecture 4: Retrieval-Augmented Generation (RAG)

LLMs sometimes produce answers that sound correct but are wrong. This problem is called hallucination. Retrieval-Augmented Generation (RAG) reduces hallucination by giving the model real documents to read before answering.

RAG pipeline:
1. Split documents into small chunks.
2. Convert each chunk into an embedding.
3. Store the embeddings in a vector database such as ChromaDB, FAISS or Pinecone.
4. When a user asks a question, convert the question into an embedding.
5. Retrieve the most similar chunks (top-k) from the vector database.
6. Send the question together with the retrieved chunks to the LLM, which writes the final answer.

A vector database is different from a normal SQL database. A SQL database finds rows by exact values, while a vector database finds items by similarity of meaning. Semantic search is search based on meaning rather than exact keywords.

Figure 4.1: The RAG Pipeline



The question is embedded and compared with every chunk using cosine similarity.